

Soohwan Kim — Future CFD Leader

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Working Experiences

SIMULIA R&D Quality Engineer <i>Tested fluid and conjugate heat transfer simulations, and incorporated into regression suites</i> <i>Tested parallel I/O and GPU acceleration for HPC environments</i>	Dassault Systèmes 2025
Multiphase Applied CFD Engineer <i>Validated LBM-based phase-field multiphase flow solver</i> <i>Published a journal paper based on internship achievements</i>	Dassault Systèmes 2023
President <i>Led Korean community of approximately 400 graduate students</i> <i>Managed the official website and hosting networking events</i>	GT Korean Student Association 2021–Present
Graduate Research Assistant <i>Researched aerosol collection and pathogen detection by wet cyclone</i>	Georgia Institute of Technology 2021
Graduate Teaching Assistant <i>Taught fundamentals of fluid mechanics</i>	Georgia Institute of Technology 2022
Home Appliance Engineer <i>Designed wall-mounted air purifier combined with mask sterilizer</i>	Samsung Electronics 2021
Environmental Plant Engineer <i>Researched turbo blowers for water treatment plant</i>	Samsung Engineering 2020
Undergraduate Teaching Assistant <i>Teaching fundamentals of dynamics and robotics</i>	Seoul National University 2020
Translator & Squad Leader <i>Translated classified military intelligence, developed an RPA to translate Korean into English, led a squad comprised of 19 underclass soldiers</i>	Republic of Korea Army 2018–2019

Education

Georgia Institute of Technology <i>Doctor of Science in Mechanical Engineering (GPA: 4.0/4.0)</i> focus: CFD and biological fluid mechanics	United States 2021–Present
Georgia Institute of Technology <i>Master of Science in Computational Science and Engineering (GPA: 4.0/4.0)</i> focus: Machine learning-based data analytics and surrogate modeling	United States 2024–Present
Seoul National University <i>Bachelor of Science in Mechanical Engineering (GPA: 3.9/4.3, Summa Cum Laude)</i>	Republic of Korea 2015–2021

Patents

Recirculation System for Aerosol Collectors using Liquid Collection Buffer <i>US Patent Application</i>	2025
Wet Cyclone with Reduced Stagnation Zones for Improved Liquid Transport and Recovery <i>US Patent Application (pending)</i>	2026

Journal Publications

Fluid Recovery and Efficiency Curve of Wetted Wall Cyclone using LBM based Particle Model

Computers & Fluids (in prep.)

Meniscus Profiling of Fly Fishing Lures by Optical Diffraction

Integrative and Comparative Biology (in prep.)

Self-Righting of Frog Eggs

Physics of Fluids. (in prep.)

When Animals Turn Inside Out: The Eversion of Bloodworms

Bioinspiration & Biomimetics, 2026

Predicting Mosquito Flight Behavior using Bayesian Dynamical Systems Learning

Science Advances, 2026

EGFET-based detection of airborne E. coli using a recirculating wetted wall cyclone collection

Sensors and Actuators B: Chemical, 2025

Capillary Flow Simulation with the Phase-field-based Lattice Boltzmann Solver

Physics of Fluids, 2025

Flotation of California Blackworms and Other Hyponeuston

Interface Focus, 2025 (Cover Image)

Enhanced Wet Grip with North American River Otter Paws

Annals of the New York Academy of Sciences, 2024

Canine-inspired Unidirectional Flows for Improving Memory Effects in Machine Olfaction

Integrative and Comparative Biology, 2023

Onggi's Permeability to Carbon Dioxide Accelerates Kimchi Fermentation

Royal Society Interface, 2023 (Cover Image)

Media Coverage

The Best Kimchi Is Made in Earthenware Pots. Science Reveals Why

The Washington Post

2023

Watch superstretchy bloodworms turn themselves inside out

Science News

2025

Celebrating Korean Culture At The American Museum Of Natural History

Forbes

2025

New model predicts how mosquitoes will fly

MIT News

2026

Awards & Fellowships

Departmental Topping Fellowship of Best Graduate Mentor

George W. Woodruff School of Mechanical Engineering

2024

Best Mentor Award

Nakatani Foundation of Research and International Researches for Students

2024

KSEA-KUSCO Graduate Scholarship

Korean-American Scientists and Engineers Association

2023

Presidential Science Scholarship

Korea Student Aid Foundation

2015–2020

Professional Skills

COMSOL, OpenFoam, Star-CCM+, Ansys Fluent/Icepak, HyperWorks, PowerFLOW, ParaView

(computational tools for CFD, aerodynamics, and heat transfer)

Solidworks, Siemens NX, FDM/SLA Printing, LabVIEW, Arduino, Ansys ICEM, HyperMesh, ABAQUS

(design/prototyping/experiment, CAD, meshing, and FEM tools)

C++, Python, Julia, MATLAB, Pytorch (coupled with OpenFOAM), OpenCL, and GPU computing

(programming languages with AI tools and parallel computing, ML-based feature extraction and surrogate model)